

Topic quiz: Power and efficiency

IBDP Physics SL · Grade 11 · A.3 Power and efficiency · 10 multiple-choice questions

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Name: _____

Class: _____

Date: _____

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use $g = 9.81 \text{ m s}^{-2}$ and show any working next to the question.

Check yourself online: the same quiz with instant feedback and explanations —

<https://biba0112.github.io/physics-sims/quiz-power-efficiency/>



1	2	3	4	5	6	7	8	9	10

Question 1

A 60.0 kg student runs up stairs with a vertical height of 3.0 m in 4.0 s. What is the useful power output?

- (A) 45.0 W (B) 441 W
(C) 1770 W (D) 7060 W

Question 2

A car travels at a constant 25 m s^{-1} against a total resistive force of 800 N. What power does the engine deliver to the wheels?

- (A) 0 W, because the resultant force is zero
(B) 32 W
(C) 800 W
(D) 20 kW

Question 3

A motor raises an 800 kg lift at a constant speed of 1.5 m s^{-1} . What is the useful power output of the motor?

- (A) 0.90 kW (B) 1.2 kW
(C) 5.2 kW (D) 11.8 kW

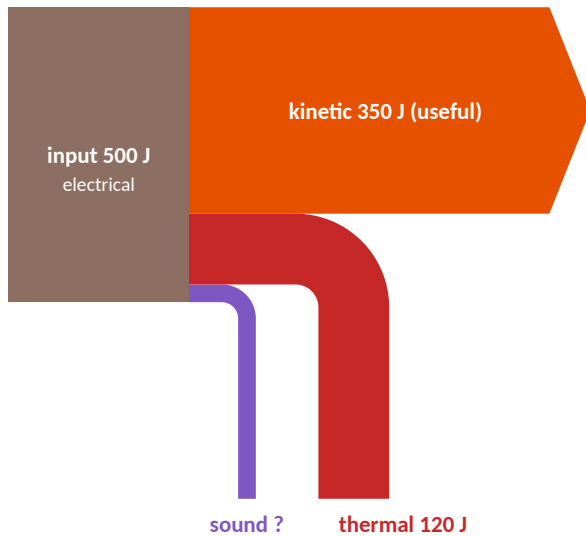
Question 4

An electric motor uses 3.5 kJ of energy to lift a 150 N load through 7.2 m. What is its efficiency?

- (A) 3.1 % (B) 31 %
(C) 69 % (D) 324 %

Question 5

The Sankey diagram shows the energy transfers in an electric motor every second. What is the missing sound energy and the efficiency of the motor?



- | | |
|----------------------------------|---------------------------------|
| (A) Sound 30 J, efficiency 70 % | (B) Sound 30 J, efficiency 30 % |
| (C) Sound 150 J, efficiency 70 % | (D) Sound 30 J, efficiency 74 % |
-

Question 6

A ball is dropped from 1.20 m and bounces back up to 0.84 m. What percentage of its energy does it keep after the bounce?

- | | |
|----------|-----------|
| (A) 30 % | (B) 70 % |
| (C) 84 % | (D) 143 % |
-

Question 7

Petrol has an energy density of 46 MJ kg^{-1} . A car carries 40 kg of petrol, its engine is 25 % efficient, and at 25 m s^{-1} it needs 20 kW of useful power. About how far can it travel on this petrol?

- | | |
|------------|-------------|
| (A) 23 km | (B) 144 km |
| (C) 575 km | (D) 2300 km |
-

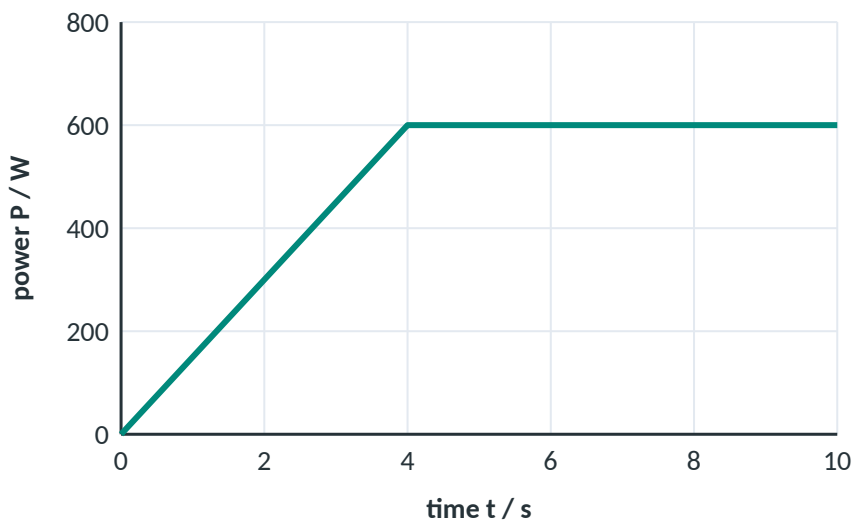
Question 8

Why can no machine have an efficiency greater than 100 %?

- | |
|--|
| (A) The useful energy out cannot be more than the total energy in, because energy is conserved |
| (B) Machines always need some input power |
| (C) Friction always wastes exactly half the energy |
| (D) Power is always lost faster than energy |

Question 9

The graph shows the power output of a motor. How much energy does it transfer in the 10 s?



- (A) 60 J (B) 3600 J
(C) 4800 J (D) 6000 J

Question 10

A 1200 kg car drives up a 5.0° slope at a constant 15 m s^{-1} against a resistive force of 600 N. What is the power output of the engine?

- (A) 9.0 kW (B) 15.4 kW
(C) 24.4 kW (D) 185 kW

End of quiz • Score: _____ / 10